

Predictive Weight Optimization: A Machine Learning Approach to Maximizing Expected Value in Men’s March Madness Brackets

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Abstract

We model the NCAA Division I Men’s Basketball Tournament as a 63-game probability tree, and formulate the problem of choosing a 64-team bracket which maximizes expected ESPN Tournament Challenge points. To do so, we (i) fit a logistic-regression win-probability model to fifteen seasons (2010-2026) of KenPom efficiency data and tournament results (excluding 2020) and (ii) vectorize a Monte Carlo bracket simulator. We compare 3 deterministic strategies: chalk-by-seed, chalk-by-model and EV-max. The full model achieves an 83.0% accuracy and 0.410 log-loss on leave-one-season-out cross-validation. Backtesting 15 seasons shows that the model bracket wins 13/15 matchups over the seed-chalk; its realized score has 428 more points (52% relative improvement). In 2026, the model bracket scores 1,570 points to the seed-chalk’s 1,000 points, and selects Michigan as champion with a 67.2% probability.

1 Problem Description

In March, sixty-eight Division I men’s basketball teams participate in a single-elimination tournament (sixty-four teams after the play-in “First Four”) to determine the national champion. Tens of millions of fans complete a *bracket*, their predictions of the winners of all sixty-three games before the tournament even starts. ESPN’s Tournament Challenge alone collected over 26.6 million brackets in 2026 [9], a record for the fourth consecutive year.

Picking a good bracket is difficult. In a single-elimination tournament, even the pre-tournament favorite has probability less than 25% [1] of winning all six games. The scoring is also non-linear: ESPN assigns 10, 20, 40, 80, 160, and 320 points, respectively, to the R64, R32, S16, E8, F4, and championship round games, so picking the champion is estimated to be worth 32 first round wins. Finally, the decision space is also very large: there are some $2^{63} \approx 9.2 \times 10^{18}$ possible brackets.

Note that per-game accuracy never maximizes expected total score since later rounds are worth far more points. This is a game where one does not maximize per-game accuracy, but rather expected total score.

Goal. We first build a calibrated win-probability model based on previous performance metrics, and then derive a rule for filling in the bracket that maximizes expected ESPN score under that model. We validate this model using all available NCAA tournaments from 2010 to 2026 with complete data.

2 Simplifications

An entire model would include injuries, travel, officiating, the joint distribution of pool-entrant strategies, and prize structure. We make typical simplifications [5, 7].

- **First Four excluded.** In modeling the 64-team bracket beginning with R64, we treat each First Four winner as their corresponding seed.
- **Static, season-end features.** All KenPom features are static end-of-regular-season values; brackets are entered before any tournament games are played.
- **Independent games.** Game results are conditionally independent given team attributes.
- **Symmetric matchup framing.** Since each matchup is the signed difference of Team A’s features from Team B’s, $P(A \text{ beats } B) = 1 - P(B \text{ beats } A)$ by construction.
- **Momentum from game logs, with a fallback proxy.** Late-season momentum is a decay-weighted win rate over a team’s last 10 pre-tournament games (decay 0.85); calculated from per-game schedule data scraped from Sports-Reference [2]. For team-seasons without game-log coverage we fall back to an AdjEM $\times$ Luck proxy.
- **Public-pick data is single-year.** ESPN’s “Who Picked Whom” data was only available for 2019, meaning pool-relative analysis is theoretical (§3.5) rather than backtested.
- **2020 dropped from the backtest** (no tournament due to COVID).

3 Mathematical Model

3.1 Win-probability oracle

Let $\mathcal{T}_y$ denote the 64 tournament teams in year y . For each team $t \in \mathcal{T}_y$, we create 11 features by calculating 10 KenPom statistics: Adjusted Efficiency Margin (AdjEM), Adjusted Offensive Efficiency (AdjO), Adjusted Defensive Efficiency (AdjD), Adjusted Tempo (AdjT), Luck, two strength-of-schedule statistics (SOS-AdjEM and NCSOS-AdjEM), the decay-weighted Momentum signal of §2, the Four Factors statistics offensive eFG% (Off-eFG%) and defensive eFG% (Def-eFG%), and a seed. For each ordered pair (a, b) of teams, we define the signed feature difference

$$\Delta_{ab} = \mathbf{x}_y(a) - \mathbf{x}_y(b) \in \mathbb{R}^{11}, \quad \Delta_{ab} = -\Delta_{ba}. \quad (1)$$

We model the winning probability as

$$P_\theta(a \text{ beats } b) = \sigma(\theta_0 + \boldsymbol{\theta}^\top \Delta_{ab}) = \frac{1}{1 + \exp(-(\theta_0 + \boldsymbol{\theta}^\top \Delta_{ab}))}, \quad (2)$$

where σ is the logistic function. It has the property that reversing the pair reverses Δ , so $P_\theta(a, b) + P_\theta(b, a) = 1$. Following Lopez and Matthews [7] in this domain, we treat each game as a binary classification problem minimizing the log-loss.

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)], \quad (3)$$

where $\hat{p}_i$ is the model’s predicted probability of winning matchup i . Log-loss is the scoring rule of choice when the downstream task depends on calibrated probabilities.

3.2 The bracket as a probability tree

The bracket is a balanced binary tree of depth six, with 63 internal nodes (games) and 64 leaves (teams), four regions of 16 teams each, and a Final Four. Given P_θ it induces a joint distribution on the random vector $(R_t)_{t \in \mathcal{T}_y}$ where $R_t \in \{0, \dots, 6\}$ is the maximum round reached by team t . This distribution is not factored because each team’s path depends on the opponents it matches up against, so we estimate it by Monte Carlo (§4.3).

3.3 ESPN scoring and the EV objective

A bracket entry $\pi = (\pi_R)_R$ indicates for each round R which teams are picked to advance. The ESPN scoring rule is

$$S(\pi, R) = \sum_R w_R \cdot \#\{t \in \pi_R : R_t \geq \text{idx}(R)\}, \quad w = (10, 20, 40, 80, 160, 320), \quad (4)$$

with a maximum score of 1,920. Define the marginal reach probability $\rho_R(t) := \Pr[R_t \geq \text{idx}(R)]$. The expected score is

$$\text{EV}(\pi) = \sum_R w_R \sum_{t \in \pi_R} \rho_R(t). \quad (5)$$

By the linearity of expectation, the EV-maximizing entry under a fixed model is the closed-form greedy rule: at every game, pick whichever reachable team has the higher ρ_R . In particular, there is no need to search over the 2^{63} bracket space. This is a key methodological observation of the project.

3.4 Bracket strategies

We enumerate three deterministic pick strategies:

1. **chalk_seed**. Pick the lower-seeded team in every game. Baseline.
2. **chalk_model**. At every game, choose the action $\arg \max_c P_\theta(c \text{ wins})$.
3. **ev_max**. For each game in round R compute $\arg \max_c \rho_R(c)$.

For a calibrated symmetric model, **chalk_model** and **ev_max** coincide because respective per-game $\arg \max$ s / reach-probability $\arg \max$ s agree about which team to bet on. We confirm this empirically in §5.4.

3.5 The pool-percentile objective (theoretical extension)

EV is not the right target; after Metrick [8], we define a field of M competitor brackets sampled from the public-pick distribution and a target percentile α . The pool-finish probability is

$$\text{PFP}_\alpha(\pi) = \Pr \left[\frac{1}{M} \sum_{i=1}^M \mathbf{1}[S(\pi, R) > S(\pi^{(i)}, R)] \geq \alpha \right]. \quad (6)$$

A natural heuristic is, for every game in $R \geq \text{S16}$, choose the team c maximizing $\rho_R(c)[1 - q_R(c)]$, where $q_R(c)$ is the public pick rate. This leverages the model where it disagrees with the crowd. Since we only have one year of public-pick data, we do not include this approach in the backtest below.

4 Solution

4.1 Training the win-probability model

The matchup dataset has **2,014 rows** covering every actual tournament game played between 2010–2026 [2] (excluding 2020), with mirrored orderings yielding exact 50/50 class balance across the 1,007 games. In each LOSO training fold, we standardized the features. We validated models by leave-one-season-out cross-validation, as it better resembles the forecasting problem of training on past seasons to predict a future tournament.

Model	Log-loss	Std	Accuracy	Brier
Logistic, seed only	0.572	0.046	70.5%	0.195
Logistic, full features	0.410	0.090	83.0%	0.129
XGBoost [4]	0.434	0.094	81.6%	0.135
CatBoost [6]	0.442	0.092	80.8%	0.137
Random Forest [3]	0.471	0.060	78.3%	0.153
LightGBM [5]	0.518	0.140	81.2%	0.145

Table 1: Model comparison under leave-one-season-out cross-validation.

Full logistic regression achieves the best log-loss and Brier score, and so is used for downstream tasks. While tree ensembles achieve similar accuracy, they are worse at probability calibration, exactly the property we need to compute EV. Adding the ten KenPom features to the seed-only baseline reduces log-loss by 28%. Given the small number of $\sim 2,000$ training rows, tree ensembles may not be able to capture the non-linearities successfully. The LightGBM high variance across folds also suggests the model is overfitting on the small per-season slices. In the reliability diagram in Figure 1, the predicted probabilities are well-calibrated, meaning that predictions closely track observed frequencies across the probability range.

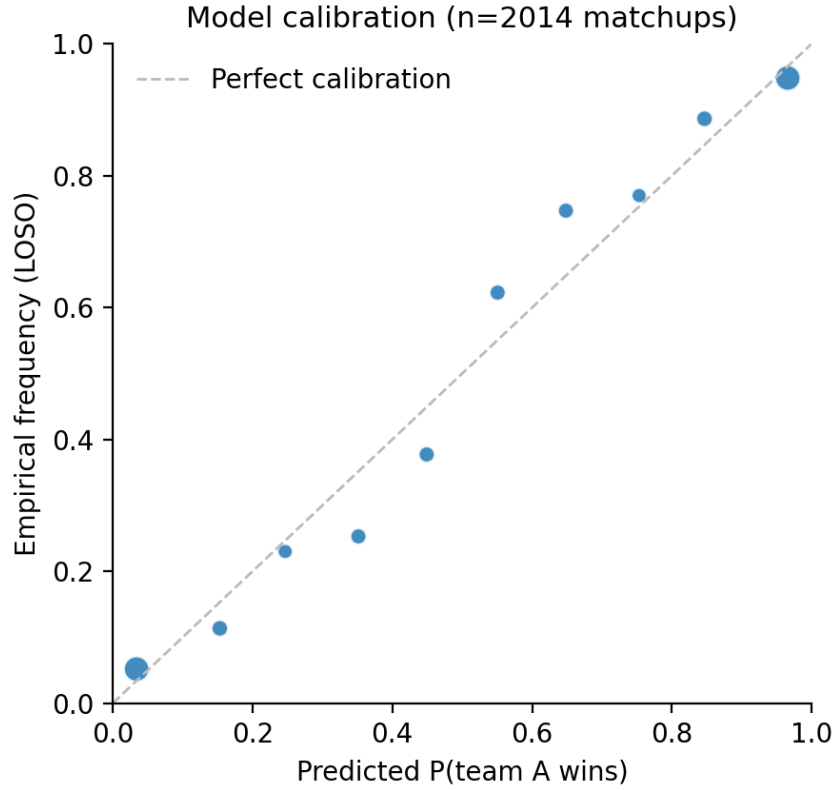


Figure 1: Reliability (calibration) diagram for the full logistic model on 2,014 LOSO predictions across 2010–2026.

4.2 Fitted coefficients

The features are standardized, making coefficients interpretable as effect sizes on the response.

Feature	$\hat{\theta}$	Feature	$\hat{\theta}$
d_Seed	+2.60	d_Momentum	−0.65
d_AdjEM	+1.83	d_OffEFG	−0.11
d_AdjO	+1.62	d_NCSOS-AdjEM	−0.06
d_AdjD	−1.42	d_DefEFG	+0.02
d_SOS-AdjEM	+1.36	d_AdjT	+0.02
d_Luck	+1.10		

Table 2: Standardized logistic regression coefficients.

These are sensible. Better teams (lower seeds) are more efficient (higher offensive rating, lower defensive rating), and play harder schedules, but tempo makes no difference. The Four Factors coefficients are all near zero due to multicollinearity with the adjusted efficiencies they are used to form. The Momentum coefficient is slightly negative because the decay-weighted

winning percentage is correlated with efficiencies already in the model, and teams that get hot late in the year tend to regress during the tournament. Round-by-round fits (Figure 2) show the Momentum coefficient becoming increasingly negative through S16 and attenuating thereafter, with Seed and AdjEM as the primary contributors.

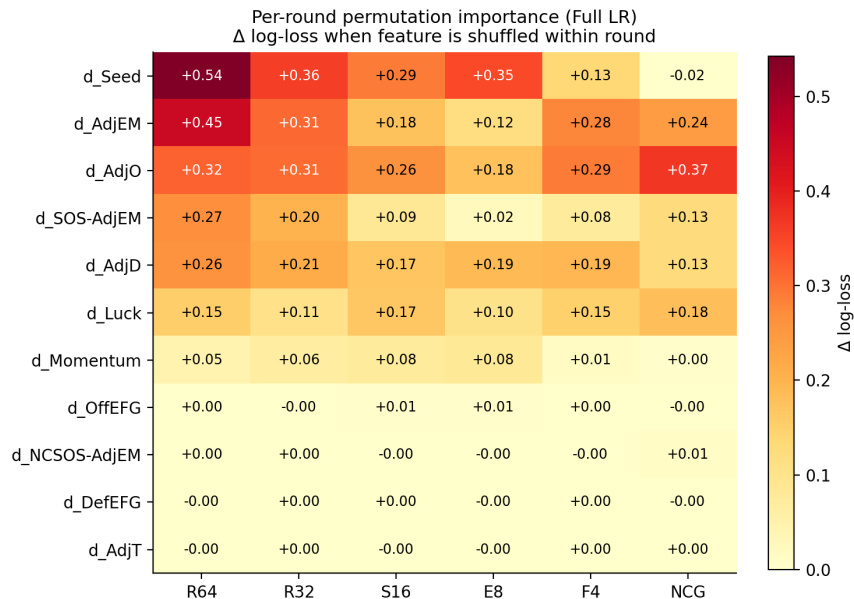


Figure 2: Round-by-round standardized coefficient magnitudes for separately fit logistic regressions.

4.3 Vectorized Monte Carlo simulator

Instead of running the 63 games per replicate through the model, we build the 64×64 matrix $P_{ij} = P_{\theta}(i \text{ beats } j)$ once per season, and during each replicate sample 63 Bernoulli draws against the precomputed lookup tables. This brings the per-replicate complexity to $O(n_{\text{sims}} \cdot 63)$, with a one-time precomputation cost of $O(64^2)$. With $n_{\text{sims}} = 10,000$, the simulated results are computed in seconds, with the per-team qualifying probabilities having a standard error of less than one percentage point.

5 Results

5.1 Historical backtest (2010–2026, $n = 15$)

For each season, we train the model on all other seasons (strict LOSO, no leakage), yielding the chalk_seed and chalk_model brackets that we would compare against the actual outcome. The full season-by-season comparison is shown in Figure 3. Summary statistics:

- The model bracket beats the seed in **13 out of 15 tournaments** (binomial $p \approx 0.004$).
- Mean score: chalk_seed **827**, chalk_model **1,255**, $\Delta = +428$ (52% relative improvement).

- The two losses are 2017 (a chalky North Carolina–Gonzaga championship) and 2025 (an exceptionally chalky tournament where seed-chalk scored 1,730).
- The model never *expects* chalk_seed to outperform: $\mathbb{E}[\text{model}] > \mathbb{E}[\text{seed}]$ in 15 of 15 seasons.

A useful internal check: the model’s mean expected score over seasons (1,250.5) closely matches the model’s realized mean score (1,255.3), indicating no systematic bias in the estimated probabilities.

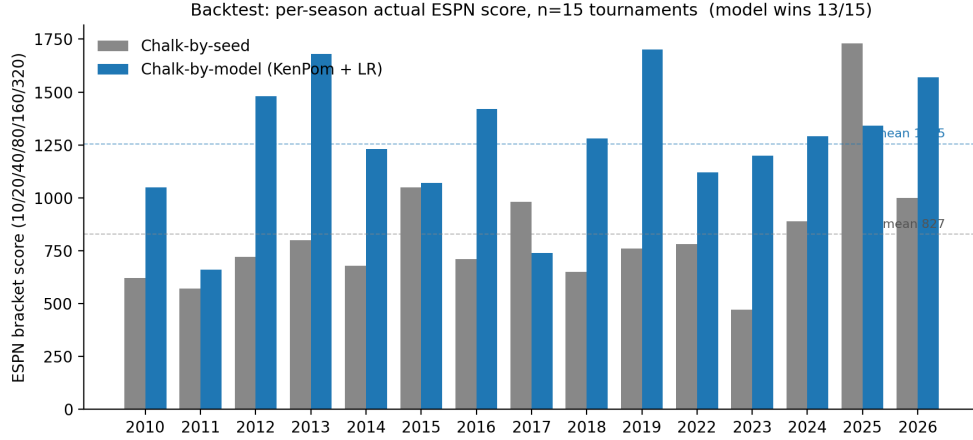


Figure 3: Per-season actual ESPN score, chalk-by-seed vs. chalk-by-model, 2010–2026 (excluding 2020).

5.2 The 2026 tournament — predictions

The model projected a heavy favorite in Michigan. Champion probabilities for the top teams (see full reach-probability heatmap in Figure 4):

Team	Seed	$P(\text{Champion})$	$P(\text{F4})$
Michigan	1	0.672	0.970
Arizona	1	0.253	0.914
Duke	1	0.069	0.839
Illinois	3	0.003	0.434
Connecticut	2	0.000	0.101

Table 3: Top 2026 reach probabilities under the trained model.

A 67.2% champion probability is unusually concentrated for any single team. The model miss on the eventual runner-up Connecticut, who had near-zero champion probability, is partially compensated for by the Monte Carlo simulation, which considers all possible tournament paths.

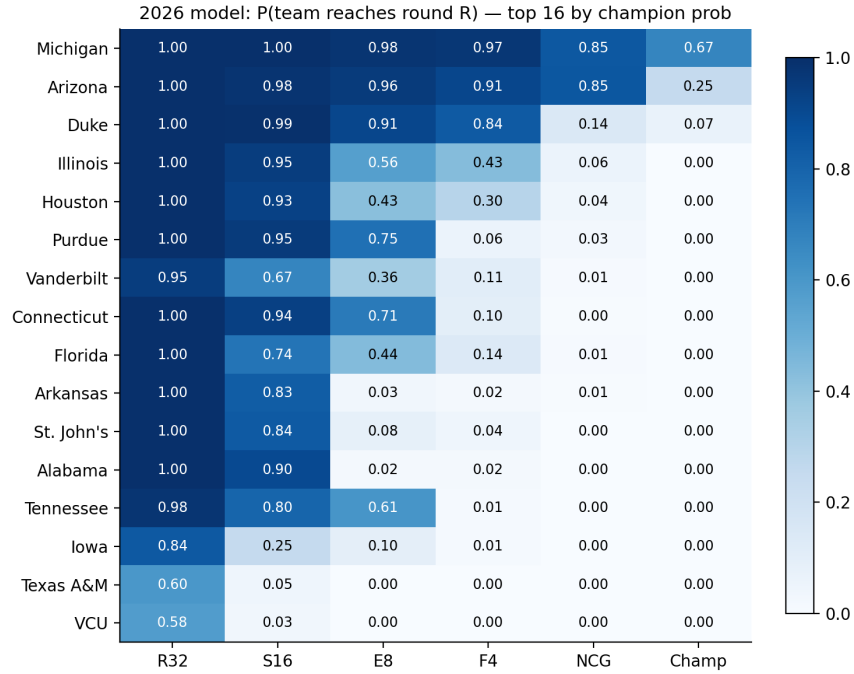


Figure 4: 2026 reach probability heatmap for the top 16 teams by champion probability.

5.3 The 2026 tournament — outcome and strategy scores

Michigan won the championship, defeating Connecticut 69-63. Upsets of the tournament included Iowa (9) over Florida (1), Texas (11) over Gonzaga (3), and Connecticut (2) defeating Duke (1) in the E8.

Strategy	Total	R64	R32	S16	E8	F4	NCG
chalk_seed	1,000	240	240	200	160	160	0
chalk_model	1,570	310	260	280	240	160	320
ev_max	1,570	310	260	280	240	160	320

Table 4: 2026 strategy scores broken out by round.

The models both decisively outperform chalk_seed (+570 each), correctly picking Michigan as the champion. chalk_model and ev_max make the same predictions for all 63 games, confirming the closed-form equivalence shown in §3.3.

5.4 Distributional analysis

In 10,000 simulated tournaments using the 2026 model:

Strategy	Mean	SD	p10	Median	p90
chalk_seed	1,301	191	1,060	1,290	1,590
chalk_model	1,528	222	1,210	1,600	1,780
ev_max	1,528	222	1,210	1,600	1,780

Table 5: Score distributions for each strategy across 10,000 simulated 2026 tournaments.

The model strategies also have a **+227-point expected score advantage** and slightly larger variance (Figure 5). The larger variance is by design: concentrating our picks on a high-probability champion increases both the mean and the right-tail of the score distribution.

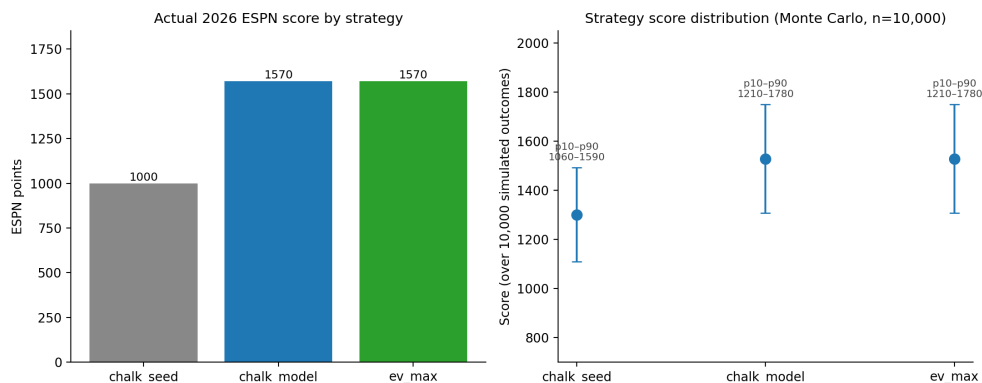


Figure 5: 2026 strategy comparison. Left: realized ESPN score. Right: distribution over 10,000 simulated outcomes.

6 Discussion

Three general points arise. First, calibration is more important than marginal accuracy. The full LR is still slightly better than tree ensembles with respect to log-loss and Brier score, both relevant for expected value decisions. When decisions depend on probability *magnitudes*, a well-calibrated linear model can outperform a higher-accuracy non-linear one—this is the gap between predicting winners and estimating probabilities you can actually compute expectations against.

Nonlinear scoring increases model confidence. In 2026, the model’s strong belief in Michigan produces a champion distribution sharply concentrated on Michigan and a well-performing bracket. The backtest confirms this: the difference between the model and seed reflects tournament "chalkiness".

The EV optimization collapses to a greedy rule. What looks like a 2^{63} -element search problem is solved by inspection given a fixed model: at every node, pick the team with higher reach probability. The hard work has been pushed into the calibration of the model—exactly where modeling effort should sit. The downside of this greediness is that EV maximization and pool-percentile maximization are not the same. The leverage strategy from §3.5 provides a tractable theoretical bridge, and future multi-year public-pick data could sharpen it.

7 Improvements

The key limitations relate to what data is available: only single-year (2019) public-pick data is available for pool-percentile optimization. Additionally, late-season injuries, suspensions, and styles of play for potential matchups could not be included in these ratings. The small (63 games/year $\times$ 15 year) dataset prevents observing complicated patterns (but these are properties of the tournament data, not limitations of the approach).

Extensions include game-log scraping for all KenPom-rated teams (enabling conference-tournament margin of victory and obviating the Momentum proxy), multi-year public-pick scraping to back-test the §3.5 leverage heuristic, PFP optimization via simulated annealing directly on the bracket space, directly learning θ to maximize expected bracket EV instead of minimizing the log-loss, via policy-gradient or differentiable simulation, and round-specific feature weights, motivated by the round-by-round fits reported in §4.2, which suggest feature importance varies as the tournament progresses.

8 Conclusions

We built an end-to-end NCAA tournament prediction pipeline comprised of data collection, a calibrated model of win probability, a vectorized Monte Carlo simulation, and three deterministic bracket-building strategies. In backtesting, the model bracket scores 52% more ESPN points than the seed-chalk bracket on average and wins 13 out of 15 past tournaments. For 2026, the model bracket scored 1,570 to 1,000 for seed-chalk, predicting Michigan would win with a success probability of 67.2%.

Two methodological observations: since the EV objective decomposes by linearity of expectation, the entry that is EV-maximizing under the fixed model gives a *closed-form greedy* solution. Most “optimization” formulations of bracket selection ignore that exponential search is not necessary. A simple example is that `chalk_model` and `ev_max` agree on every decision for all 63 2026 games. Second, the gap between EV maximization and pool-percentile maximization is real and persistent, and the leverage heuristic of §3.5 is a tractable theoretical bridge for future work.

In a class on discrete modeling and Monte Carlo simulation, this project viewed the bracket as a depth-six binary tree, the win-probability model as the per-edge transition kernel, and Monte Carlo methods as the only way to calculate the resulting joint distribution.

Acknowledgments

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